Deep Learning Project

A Deep Learning Approach to Artist Classification on WikiArt

Group 21

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Introduction

Attributing a painting to its author from pixels alone is a genuinely difficult problem. Unlike object classification, where the target is a discrete, well-defined shape, artistic style is spread across many signals at once like texture, colour choice, how light and dark areas are distributed and how an artist tends to arrange a scene. A model that learns only one of these signals will fail on any artist whose work spans different subjects or periods.

This work explores the application of deep learning to artist classification on a 23-class subset of WikiArt, a large-scale dataset of art paintings spanning diverse Western artistic traditions. The study investigates the role of data augmentation designed specifically for paintings, the effectiveness of progressive fine-tuning of pretrained backbones, and the benefit of model ensembling for a task where artistic styles can look very similar across artists and there is a relatively small number of training images per artist.

Exploratory Data Analysis

As with any data-driven project, the first phase consisted of examining the dataset to understand which aspects needed treatment before modelling. One clear issue was that number of images per artist class was uneven. Some artists are represented by a larger number of works while others have fewer samples, while the majority had around 300 to 500 paintings, Nicholas Roerich and Vincent van Gogh stand out with more than 1,200 paintings each. This imbalance affected model development and evaluation choices.

Each artist has a distinctive style, whether in subject matter, composition, or colour choices, which shaped data augmentation decisions. Two artists stood out, Albrecht Dürer and Gustave Doré paint almost entirely in grayscale. And significant differences in colour warmth were noticed across artists. These observations led to a conservative approach to augmentation to avoid removing unique artist style features.

Duplicate detection was performed by converting images to perceptual hashes and comparing them using Hamming distances. A threshold of 10 was selected after finding that higher values began producing false positives, for example by confusing portraits and self-portraits across artists, making manual verification impractical. No corrupted images were found, though some Nicholas Roerich paintings looked low information due to his pencil-on-white-paper style, which was correctly identified as intentional and retained. Potential outliers, such as paintings which looked unusual from an artist's

typical style, were deliberately kept, as a good model should handle stylistic evolution within an artist's career.

Data Preparation

After removing duplicates, the dataset had 13,172 paintings across 23 classes. Images were resized to *224 x 224* pixels to ensure compatibility with some ImageNet-pretrained models, accepting some loss of detail in exchange for faster computational efficiency. One-hot encoding was used over integer labels to ensure direct compatibility with categorical cross-entropy loss, label smoothing, and the macro F1 metric. The data was split 70/30 into training and held-out sets, with the held-out portion further divided into validation and test. The validation set was used for model selection throughout all experiments, while the test set was only looked at once for final evaluation.

The largest class (Van Gogh, 1,322 images) had approximately four times more paintings than the smallest (Dalí, 336 images). To handle this, balanced class weighting was applied, scaling each sample's loss contribution inversely proportional to its class frequency. This was preferred over oversampling, which would introduce duplicate images and risk overfitting minority classes, and over undersampling, which would discard useful training data from well-represented artists.

Due to this class imbalance, macro F1 was chosen as the main evaluation metric. Unlike accuracy, which is dominated by frequent classes, macro F1 computes the mean of precision and recall separately for each artist and averages them with equal weight regardless of support, making it a more honest measure of performance across all 23 classes.

Model Architecture and Training

CNN from Scratch

A medium-sized CNN was trained from scratch to set a performance baseline. The architecture consists of four convolutional blocks followed by a classification layer. Each block applies two Conv2D layers with *3 x 3* kernels, batch normalisation, and ReLU activation, followed by Dropout. The first three blocks use *2 × 2* Max- Pooling to reduce image size, while the fourth uses Global Average Pooling to produce a fixed-length feature vector. Filter counts double at each block (32 to 64 to 128 to 256) and L2 regularisation with $\lambda = 10^{-4}$ was applied throughout. The model was trained with Adam optimaser at $lr = 10^{-3}$, categorical cross-entropy with label smoothing of 0.1, early stopping (patience=8) and ReduceLROnPlateau (factor=0.5, patience=4), for up to 60 epochs.

Augmentation strategy for painting classification requires careful consideration of which transformations preserve discriminative features. Vertical flipping is ruled out because paintings have a canonical orientation determined by gravity, horizon lines, and figure posture. Aggressive rotation and zoom were excluded as they distort compositional and perspective cues that differ across artists. Low-intensity adjustments (±10%) were safer and offer sufficient variability to regularise without destroying colour signals. Three strategies were evaluated, identity, geometric (horizontal flip, rotation, zoom), and photometric (horizontal flip, brightness, contrast), with results summarised in Table 1.

The scratch CNN ceiling of 0.633 val F1 establishes the baseline for transfer learning experiments. The consistently small train/val gap across all variants indicates the bottleneck is representational capacity rather than overfitting, the model lacks the depth and pretrained texture detectors necessary to separate stylistically similar artists such as the Impressionist cluster.

EfficientNetV2S

EfficientNetV2S was selected as the primary pretrained model because its design balances depth, detail, and image resolution in a way that works well for tasks requiring fine visual distinctions. It was originally pretrained on ImageNet-21k with 21M parameters, giving it a strong foundation that still runs efficiently on a standard research GPU. The same brightness and contrast adjustments from our earlier experiments, and left out rotation and zooming since they distort how a painting is composed which is one of the key things that sets artists apart.

The model was trained in three phases, gradually unlocking more of its layers over time. This was done to adapt it to art paintings without erasing the general visual knowledge it had already learned from its original training data. In Phase 1, the backbone was fully frozen and only the classification head was trained at $lr = 10^{-3}$, since at this point the deep pretrained layers was kept fixed, the classification head was the only part that needed to learn. The head consists of a Dense layer with 256 units, Dropout (0.3), and a 23-class softmax, achieving a val F1 of 0.677. In Phase 2, block6 (layers 288 onwards) was unfrozen at $lr = 10^{-4}$. Block6 is the deepest and most task-specific block in EfficientNetV2S, containing 15 sub-blocks with Squeeze-and-Excitation attention, the layers most responsible for high-level semantic representations that require adaptation from ImageNet to painterly features. Earlier layers encoding generic low-level features such as edges and textures remained frozen. Phase 2 achieved a val F1 of 0.840. In Phase 3, block5 (layers 155 to 287) was additionally unfrozen at $lr = 5 \times 10^{-5}$, allowing the model to also adapt its mid-level layers, the ones responsible for picking up on brushstroke patterns and surface textures. This added a further +0.010, bringing the final validation F1 to 0.850.

A regularised variant with permanently frozen BatchNormalisation layers, increased Dropout (0.4), stronger L2 ($\lambda = 3 \times 10^{-4}$), and a more conservative unfreeze scope (last three sub- blocks of block6, $lr = 10^{-5}$) reduced the train/val gap from 0.150 to 0.095 at a cost of 11.3 val F1 points, The takeaway is a recurring tradeoff, the same settings that stop the model from memorising the training data also limit how well it can adapt to the new task.

MobileNetV3Large

MobileNetV3Large was selected as a complementary backbone for ensembling. With 5.4M parameters and a depthwise separable convolution architecture, it provides a different inductive bias to EfficientNetV2S's Fused-MBConv structure, making it likely to produce

different error patterns on ambiguous cases. The same head, augmentation, and frozen BatchNormalisation strategy were applied. Phase 1 achieved a val F1 of 0.713 with a gap of 0.035, the smallest phase 1 gap across all models, reflecting lower backbone capacity. Phase 2 unfroze the last three expanded convolutional blocks (layers 140 onwards) at $lr = 10^{-4}$, achieving a val F1 of 0.803. A regularised variant with a more conservative unfreeze scope (layers 155 onwards, $lr = 5 \times 10^{-5}$) reduced the gap from 0.190 to 0.153 at a cost of 3.1 val F1 points, reinforcing the same memorisation-adaptation tradeoff observed with EfficientNetV2S.

Ensemble

The two best single-model checkpoints, EfficientNetV2S (val F1 0.850) and MobileNetV3Large (val F1 0.803), were combined into an ensemble by averaging their softmax output distributions. Despite sharing the same training objective and augmentation strategy, the two models differ substantially in architecture, parameter count, and inductive bias, making them suitable ensemble partners. EfficientNetV2S uses Fused-MBConv blocks with compound scaling and learns richer but more dataset-specific representations, while MobileNetV3Large uses depthwise separable convolutions with a much smaller parameter budget. These architectural differences lead the two models to assign different confidence distributions to ambiguous cases, particularly in the Impressionist cluster where no single set of features is definitively discriminative.

Two weighting schemes were evaluated, equal weights (50/50) and a V2S-favouring scheme (60/40), to determine whether the stronger model should contribute more to the final prediction.

The equal-weight ensemble outperformed both individual models and the weighted variant, gaining +0.012 over EfficientNetV2S alone. The fact that equal weighting outperforms 60/40 is informative, the two models are making different mistakes rather than one simply being better than the other. Giving more weight to V2S reduces the diversity benefit and partially suppresses the corrective contribution of MobileNetV3Large on the cases where it disagrees correctly. The 50/50 ensemble was therefore selected as the final model.

Final Results

The final ensemble was evaluated once on the held-out test set of 1,984 images. The ensemble achieved a macro F1 of 0.860, an accuracy of 0.874, and a Cohen's Kappa of 0.867, indicating near-perfect agreement with ground truth labels beyond chance. The Matthews Correlation Coefficient of 0.867, which accounts for all cells of the confusion matrix simultaneously, corroborates this result independently. The macro ROC AUC of 0.991 confirms that the model separates each artist from the remaining 22 with near-perfect probabilistic discriminability, suggesting the learned representations are well-structured in the output probability space even in cases where the argmax prediction is incorrect.

The tight alignment between validation macro F1 (0.862) and test macro F1 (0.860) confirms that model selection on the validation set did not introduce optimistic bias, the model generalises cleanly to unseen data.

Table 1 summarises the macro F1 score on the validation set for each model, and the final ensemble result on the held-out test set.

Table 1: Validation macro F1 at best checkpoint for each model configuration, and final test set performance for the ensemble.

Model	Config	Val F1	Train F1	Gap
CNN Scratch	No augmentation	0.6330	0.6806	0.0476
CNN Scratch	Geometric aug	0.4583	0.5531	0.0948
CNN Scratch	Photometric aug	0.6269	0.6528	0.0259
EfficientNetV2S	Phase 1	0.6777	0.7173	0.0396
EfficientNetV2S	Phase 2	0.8405	0.9967	0.1562
EfficientNetV2S	Phase 3	0.8505	0.9972	0.1467
EfficientNetV2S-reg	Phase 2	0.7378	0.8328	0.095
MobileNetV3Large	Phase 1	0.7129	0.7482	0.0352
MobileNetV3Large	Phase 2	0.8031	0.9935	0.1904
MobileNetV3Large-reg	Phase 2	0.7724	0.9255	0.1531
Ensemble (50/50)	Val set	0.8622	0.9974	0.1352
Ensemble (60/40)	Val set	0.8579	0.9979	0.14
Best Ensemble (50/50)	**Test set F1 Score**	**0.8597**	—	—

Table 2: Extended metrics on the test set for the best ensemble.

Metric	Value
Macro F1	0.860
Accuracy	0.874
Cohen's Kappa	0.867
Macro ROC AUC	0.991
MCC	0.867

Analysis and Critical Discussion

Scratch vs. Transfer Learning

The performance gap between the scratch CNN (val F1 0.633) and the best indiviudal pretrained model (val F1 0.8505) confirms that ImageNet representations transfer meaningfully to the painting domain despite the substantial domain gap between natural photographs and artistic works. The scratch model, even with good regularisation and plenty of training time, simply did not have enough depth or the right low-level features to tell stylistically similar artists apart, a limit that the pretrained models cleared right from their very first training phase.

The augmentation results from the scratch model are clear: geometric changes like rotation and flipping consistently made performance worse, while small brightness and contrast adjustments were mostly harmless or slightly helpful. This makes sense because an artist's style shows up more in how they compose and structure a scene

than in the angle it is viewed from. Rotating or zooming a painting changes the physical arrangement of brushstrokes which is a core part of an artist's style in a way that brightness and contrast changes do not. The practical takeaway is that augmentation techniques designed for everyday photo classification do not translate well to art style classification.

Among the pretrained models, the regularised variants reveal a consistent tradeoff, reduced overfitting gap at a meaningful cost to val F1, suggesting the base models have sufficient capacity for this task but that fine-tuning on a small dataset inevitably leads to some degree of memorisation. The ensemble provides a slight gain over the best individual model (+0.012), confirming that EfficientNetV2S and MobileNetV3Large make sufficiently complementary errors to benefit from averaging, a result consistent with their architectural differences in inductive bias and parameter count.

Systematic Confusions

Inspection of the confusion matrix highlights some interesting failure modes. Pissarro's largest confusion is toward Monet, with 9 misclassified works attributed to him, the most concentrated single off-diagonal entry among the weaker classes. Both artists work within the same Impressionist tradition, applying colour and brushwork in ways that are difficult to distinguish even for trained observers. This is arguably a reasonable mistake, and not one that an everyday viewer would find easy to spot either.

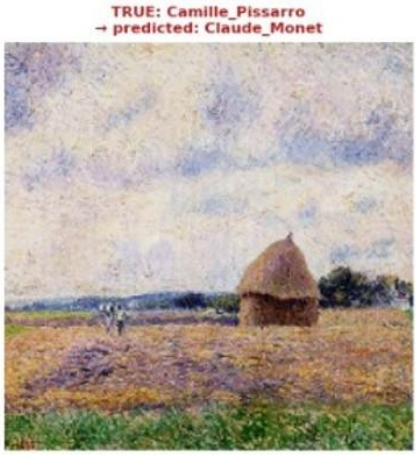


Figure 1: Pissarro predicted as Monet

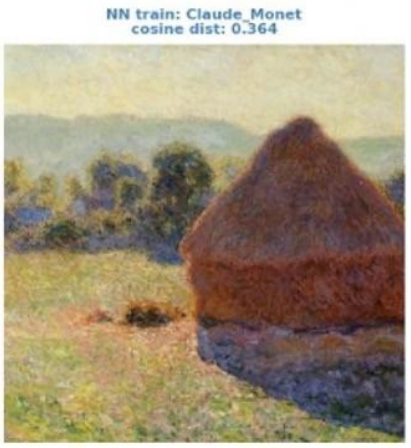


Figure 2: Example of Monet

Dalí's errors are genuinely scattered, with misclassifications spread across Cézanne (4), Monet (3), and Roerich (3), with no single dominant confusion target. Rather than reflecting a coherent stylistic proximity to any one artist, this pattern suggests the model struggles to find a consistent representational anchor for Surrealism within the feature space learned from the rest of the dataset. Dalí's work is stylistically unlike most other classes in the dataset, and the scattered errors may reflect the model defaulting to superficial cues, composition, colour temperature, subject matter, that happen to match different artists on different paintings.

Cézanne's errors are split between Renoir (3) and Van Gogh (3). The Van Gogh confusion is interpretable, both artists share a Post-Impressionist concern with visible brushwork and structural colour use, while the Renoir confusion reflects the broader

Post-Impressionist overlap in palette and subject matter.

Kustodiev's largest confusion is toward Repin (5), a fellow Russian realist. Both artists share similar subject matter, tonal palette, and painterly tradition, making this a stylistically interpretable confusion.

Roerich's largest confusion is toward Van Gogh (6), which is less immediately interpretable given the stylistic distance between the two artists. However, both share bold, saturated colours and rhythmic landscape compositions, which may have led the model to mix them up despite their very different painting styles.

Overall, the confusion matrix does not reveal a single mistake the model keeps making. Most of the errors come down to paintings by different artists simply looking too similar whether that is because they share a colour palette, paint similar subjects, or arrange their scenes in comparable ways.

Conclusions

This work investigated artist classification on a 23-class WikiArt subset through a systematic progression from scratch training to pretrained fine-tuning and ensembling. The scratch CNN established a macro F1 ceiling of 0.633, confirming that the discriminative features required for this task cannot be learned reliably without pretrained representations. Gradually adapting EfficientNetV2S and MobileNetV3Large, then combining them in a 50/50 ensemble, achieved a macro F1 of 0.860 and a ROC AUC of 0.991 on the test set. The augmentation experiments also produced a clear finding specific to paintings: geometric changes like rotation and flipping hurt performance because the way a scene is laid out is a real clue to the artist's identity, while small brightness and contrast adjustments helped regularise the model without washing out the colour choices that also distinguish artists. The primary limitation of this work is the consistently large train/val gap across all fine-tuned models, indicating substantial memorization of training paintings. Every attempt to reduce this gap through stronger regularization produced significant regressions in validation F1, suggesting that more sophisticated approaches such as Mixup augmentation would be required to resolve this tension. It is also worth noting that computational constraints shaped the experimental pipeline. Initially experiments were conducted on Google Colab, but progressive GPU quota reductions mid project required migration to Kaggle, limiting the total number of experiments that could be conducted and preventing exploration of larger backbones or higher input resolutions. The weakest classes (Dalí, Cézanne, and Kustodiev) represent a challenge that no amount of model improvement can fully solve. These artists either worked across very different styles during their careers or painted in ways that closely resemble their contemporaries. At a certain point, the problem is not necessarily a failure of the model, it's simply that artistic style itself does not always have clear boundaries.

Reproducibility. All experiments were run on Kaggle with fixed seeds (random, numpy, tensorflow: 42). Checkpoints were saved at each phase and the best validation F1 check-point is restored before evaluation.